

MATH 1ST BATCH QUESTION SOLVE

1.

(a) Idempotent matrix — definition and check

Definition. A square matrix M is **idempotent** if $M^2 = M$.

Let $A = \begin{pmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{pmatrix}$. Compute A^2 .
(A symbolic check gives)

$$A^2 = \begin{pmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{pmatrix} = A.$$

Thus $A^2 = A$, so A is idempotent.

(b) Every square matrix as sum of symmetric and skew-symmetric

Let M be any $n \times n$ matrix. Define

$$S = \frac{M + M^T}{2}, K = \frac{M - M^T}{2}.$$

Then $S^T = \frac{M^T + M}{2} = S$ (so S is symmetric) and $K^T = \frac{M^T - M}{2} = -K$ (so K is skew-symmetric). Also

$$S + K = \frac{M + M^T}{2} + \frac{M - M^T}{2} = M.$$

Hence any square matrix is the sum of a symmetric and a skew-symmetric matrix. (Uniqueness follows similarly.)

(c) Find a, b given $(A + B)^2 = A^2 + B^2$

Given

$$A = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}, B = \begin{pmatrix} a & 1 \\ b & -1 \end{pmatrix},$$

and $(A + B)^2 = A^2 + B^2$. Expanding gives

$$(A + B)^2 - A^2 - B^2 = AB + BA = 0.$$

Solving the matrix equations $AB + BA = 0$ gives

$$a = 1, b = 4.$$

2.

(a) Parameter k — number of solutions

System:

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + \left(\frac{k}{2} - 14\right)z = k + 2. \end{cases}$$

Coefficient matrix

$$M = \begin{pmatrix} 1 & 2 & -3 \\ 3 & -1 & 5 \\ 4 & 1 & \frac{k}{2} - 14 \end{pmatrix}.$$

Compute $\det(M) = -\frac{7}{2}(k - 32)$. Thus:

- If $k \neq 32$: $\det(M) \neq 0 \rightarrow$ the system has **exactly one solution**.
- If $k = 32$: $\det(M) = 0$. Substitute $k = 32$ and check ranks: the coefficient matrix has rank 2 but the augmented matrix has rank 3. So the system is **inconsistent** $\rightarrow$ **no solution**.
- There is **no** value of k that gives infinitely many solutions (for $k = 32$ it is inconsistent).

(So: unique solution for $k \neq 32$; no solution for $k = 32$.)

(b) Network flow (method; need the figure for numeric solution)

You mentioned an accompanying figure with known flow rates. I don't have that figure here, so I can't write the specific linear system or solve numeric values. Give me the figure (photo or scan) and I will:

1. Label unknown flows $x_1, x_2, \dots$
2. For each node write a balance equation: sum inflows = sum outflows.
3. Assemble the linear system $A\mathbf{x} = \mathbf{b}$ and solve (unique / infinite / none depending on rank).
4. If you specify $x_4 = 50$ and $x_6 = 0$, I will substitute and give the remaining flows and directions.

(If you want a generic example: for a node with unknowns x_1, x_2, x_3 and known inflow 10, the equation might be $x_1 + x_2 - x_3 = 10$. But please upload the actual figure for exact equations and numeric answers.)

3.

(a) Inner product, norm, distance; computations

Definitions (in $\mathbb{R}^n$).

- Euclidean inner product (dot product): for $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$,

$$u \cdot v = \sum_{i=1}^n u_i v_i.$$

- Norm (Euclidean): $\|u\| = \sqrt{u \cdot u}$.
- Distance: $d(u, v) = \|u - v\|$.

Given $u = (4, 1, 2, 3)$, $v = (0, 3, 8, -2)$, $w = (3, 1, 2, 3)$.

(i) Compute $\|3u - 5v + w\|$.

First form the vector:

$$3u - 5v + w = 3(4, 1, 2, 3) - 5(0, 3, 8, -2) + (3, 1, 2, 3).$$

A symbolic evaluation gives

$$\|3u - 5v + w\| = 3\sqrt{206}.$$

(ii) Compute $\frac{1}{\|w\|}w$ (the unit vector in direction w).

$\|w\| = \sqrt{23}$. Hence

$$\frac{1}{\|w\|} w = \frac{1}{\sqrt{23}}(3,1,2,3).$$

(iii) If $\|u + w\| = 1$ and $\|u - w\| = 5$, compute $u \cdot w$.

Use

$$\|u + w\|^2 = \|u\|^2 + \|w\|^2 + 2u \cdot w, \|u - w\|^2 = \|u\|^2 + \|w\|^2 - 2u \cdot w.$$

Subtracting: $\|u + w\|^2 - \|u - w\|^2 = 4(u \cdot w)$. So

$$u \cdot w = \frac{\|u + w\|^2 - \|u - w\|^2}{4} = \frac{1^2 - 5^2}{4} = \frac{1 - 25}{4} = -6.$$

(b) Symmetric and skew-symmetric matrices + question about $(A + B)^3$

Definitions.

- A matrix M is **symmetric** if $M^T = M$.
- A matrix N is **skew-symmetric** if $N^T = -N$.

(Examples: $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ symmetric; $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ skew-symmetric.)

Given

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{pmatrix}, B = B^T \text{ (so } B \text{ is symmetric).}$$

Since B is symmetric and A is symmetric (check: $A^T = A$ — yes), their sum $A + B$ is symmetric. For any matrix S ,

$$(S^3)^T = (S^T)^3.$$

So if $S = A + B$ is symmetric, $(A + B)^3$ is also symmetric because $((A + B)^3)^T = ((A + B)^T)^3 = (A + B)^3$. **Answer:** yes, $(A + B)^3$ is symmetric.

(c) Compute $\text{Trace}(B^T A^T + 2C^T)$ where possible

Given

$$A = \begin{pmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{pmatrix}, C = \begin{pmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{pmatrix}.$$

We form $B^T A^T$ (which is 3×3), add $2C^T$, then take trace. Computation yields

$$B^T A^T + 2C^T = \begin{pmatrix} 15 & 3 & 12 \\ 14 & 0 & 7 \\ 12 & 12 & 13 \end{pmatrix}, \text{ and Trace} = 28.$$

4.

(a) Vector space test

Given

$$V = \left\{ \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} \in M_2(\mathbb{R}) \mid a, b \in \mathbb{R} \right\}.$$

We must test whether V is a **vector space** under usual matrix addition and scalar multiplication.

Check closure under addition

Let

$$A = \begin{pmatrix} a_1 & 1 \\ 1 & b_1 \end{pmatrix}, B = \begin{pmatrix} a_2 & 1 \\ 1 & b_2 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} a_1 + a_2 & 2 \\ 2 & b_1 + b_2 \end{pmatrix}.$$

The (1,2) and (2,1) entries are **2**, not **1** $\Rightarrow A + B \notin V$.

So V is not closed under addition.

Check closure under scalar multiplication

If $k \in \mathbb{R}$,

$$kA = k \begin{pmatrix} a_1 & 1 \\ 1 & b_1 \end{pmatrix} = \begin{pmatrix} ka_1 & k \\ k & kb_1 \end{pmatrix}.$$

The (1,2) and (2,1) entries are **k**, not 1 $\Rightarrow$ **not closed under scalar multiplication**.

✓ Hence V is not a vector space, because axioms of **closure under addition** and **scalar multiplication** fail.

(b) Subspace test

Subset

$$W = \{(a, b, c, d) \in \mathbb{R}^4 \mid a \geq b\}.$$

For a subset to be a **subspace**, it must satisfy:

1. $0 \in W$
2. Closed under addition
3. Closed under scalar multiplication

Now:

- $0 = (0,0,0,0) \Rightarrow 0 \geq 0$ ✓
- If $(a_1, b_1, c_1, d_1), (a_2, b_2, c_2, d_2) \in W$,
then $a_1 \geq b_1, a_2 \geq b_2$.
Add: $a_1 + a_2 \geq b_1 + b_2$ ✓
- For scalar $k = -1$:
if $a \geq b$, then $-a \geq -b$? ✗ (This reverses inequality).

So **not closed under negative scalar multiplication**.

⊖ Therefore W is not a subspace of $\mathbb{R}^4$.

(c) Linear combination and proof

Definition

A **linear combination** of vectors $v_1, \dots, v_n$ in V is any vector of the form

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n, \alpha_i \in \mathbb{R}.$$

Proof that $W = L(S)$ is a subspace

Let $S = \{v_1, v_2, \dots, v_n\} \subseteq V$.

Define $W = L(S) = \{\sum_{i=1}^n \alpha_i v_i : \alpha_i \in \mathbb{R}\}$.

Then:

- $0 = 0v_1 + 0v_2 + \cdots + 0v_n \in W$.
- If $x = \sum \alpha_i v_i, y = \sum \beta_i v_i \in W$,
then $x + y = \sum (\alpha_i + \beta_i) v_i \in W$.
- If $k \in \mathbb{R}, kx = \sum (k\alpha_i) v_i \in W$.

✓ Hence W is a subspace of V .

Express $p(x) = 7 + 8x + 9x^2$

as a linear combination of

$$p_1(x) = 2 + x + 4x^2, p_2(x) = 1 - x + 3x^2, p_3(x) = 3 + 2x + 5x^2.$$

We want $p = \alpha p_1 + \beta p_2 + \gamma p_3$.

That means:

$$\begin{cases} 2\alpha + \beta + 3\gamma = 7 \\ \alpha - \beta + 2\gamma = 8 \\ 4\alpha + 3\beta + 5\gamma = 9 \end{cases}$$

Solving gives:

$$\alpha = 0, \beta = 13, \gamma = -2.$$

✓ Hence

$$p(x) = 0p_1(x) + 13p_2(x) - 2p_3(x).$$

(a) Direct sum

Definition

For subspaces $U, W \subseteq V$,
 $V = U \oplus W$ if

1. $V = U + W$ (every $v \in V$ can be written as $u + w$), and
 2. $U \cap W = \{0\}$.
-

Given:

$$U = \{(a, b, c) \in \mathbb{R}^3 : a = b = c\}, W = \{(0, b, c) \in \mathbb{R}^3 : b, c \in \mathbb{R}\}.$$

- Every element of U is (a, a, a) .
- Every element of W is $(0, b, c)$.

Any $(x, y, z) \in \mathbb{R}^3$ can be written as:

$$(x, y, z) = (x, x, x) + (0, y - x, z - x),$$

where $(x, x, x) \in U$, $(0, y - x, z - x) \in W$.

✓ So $U + W = \mathbb{R}^3$.

Also,

if $(a, a, a) \in U \cap W \Rightarrow (a, a, a) = (0, b, c) \Rightarrow a = 0, b = a = 0, c = a = 0 \Rightarrow$ zero vector only.

✓ Therefore $U \cap W = \{0\}$.

Thus $\mathbb{R}^3 = U \oplus W$.

(b) Basis and dimension

System:

$$\begin{cases} 2x_1 + 2x_2 - x_3 + x_5 = 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0 \\ x_1 + x_2 - 2x_3 - x_5 = 0 \\ x_3 + x_4 + x_5 = 0 \end{cases}$$

We reduce to find free variables.
Solving (simplified algebra) yields:

$$x_1 = -x_2, x_3 = -x_4 - x_5.$$

Let x_2, x_4, x_5 be free.

Then

$$(x_1, x_2, x_3, x_4, x_5) = x_2(-1, 1, 0, 0, 0) + x_4(0, 0, -1, 1, 0) + x_5(0, 0, -1, 0, 1).$$

✓ **Basis:**

$$\{(-1, 1, 0, 0, 0), (0, 0, -1, 1, 0), (0, 0, -1, 0, 1)\}.$$

✓ **Dimension = 3.**

(c) Subspaces in $\mathbb{R}^4$

$$U = \{(a_1, a_2, a_3, a_4) \mid a_2 - 2a_3 + a_4 = 0\}, W = \{(a_1, a_2, a_3, a_4) \mid a_1 = a_4, a_2 = 2a_3\}.$$

U:

Equation: $a_2 = 2a_3 - a_4$.

Let a_1, a_3, a_4 be free.

Then $a = (a_1, 2a_3 - a_4, a_3, a_4) = a_1(1, 0, 0, 0) + a_3(0, 2, 1, 0) + a_4(0, -1, 0, 1)$.

✓ **Basis(U):** $\{(1, 0, 0, 0), (0, 2, 1, 0), (0, -1, 0, 1)\}$

✓ **dim(U)=3**

W:

$a_1 = a_4, a_2 = 2a_3$.

Let a_3, a_4 free $\Rightarrow a = (a_4, 2a_3, a_3, a_4) = a_3(0, 2, 1, 0) + a_4(1, 0, 0, 1)$.

✓ **Basis(W):** $\{(0, 2, 1, 0), (1, 0, 0, 1)\}$

✓ **dim(W)=2**

$U \cap W$:

Must satisfy both:

$$a_2 = 2a_3 - a_4, a_2 = 2a_3, a_1 = a_4.$$

From first two $\Rightarrow 2a_3 = 2a_3 - a_4 \Rightarrow a_4 = 0$. Then $a_1 = 0$.

So $a = (0, 2a_3, a_3, 0) = a_3(0, 2, 1, 0)$.

✓ **Basis($U \cap W$):** $\{(0, 2, 1, 0)\}$

✓ **dim($U \cap W$)=1**

6.

(a) Transition matrices

Bases:

$$B = \{u_1 = (1, 0), u_2 = (0, 1)\}, B' = \{u'_1 = (1, 1), u'_2 = (2, 1)\}.$$

(i) Transition matrix P from $B' \rightarrow B$:

columns are coordinates of u'_1, u'_2 in basis B .

$$P = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}.$$

(ii) Transition matrix Q from $B \rightarrow B'$ is P^{-1} .

$$P^{-1} = \frac{1}{(1)(1) - (2)(1)} \begin{pmatrix} 1 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}.$$

$$\text{So } Q = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}.$$

(iii) $PQ = I$ ✓ Verified.

(b) Coordinate vector

Given $v = (2, -1, 3)$ and basis

$$S = \{v_1 = (1, 0, 0), v_2 = (2, 2, 0), v_3 = (3, 3, 3)\}.$$

We want $v = c_1v_1 + c_2v_2 + c_3v_3$.

$$c_1(1, 0, 0) + c_2(2, 2, 0) + c_3(3, 3, 3) = (2, -1, 3)$$

Gives:

$$\begin{cases} c_1 + 2c_2 + 3c_3 = 2 \\ 2c_2 + 3c_3 = -1 \\ 3c_3 = 3 \end{cases} \Rightarrow c_3 = 1, c_2 = -2, c_1 = 3.$$

✓ Coordinate vector:

$$[v]_S = (3, -2, 1).$$

7.

(a) Kernel and image

Let $T: U(F) \rightarrow V(F)$ be linear.

- **Kernel:** $\ker(T) = \{u \in U \mid T(u) = 0\}$
- **Image:** $\text{Im}(T) = \{T(u) \mid u \in U\}$

✓ **Proof that both are subspaces:**

- For image:
If $T(u_1), T(u_2) \in \text{Im}(T)$, then $T(u_1) + T(u_2) = T(u_1 + u_2) \in \text{Im}(T)$.
If $c \in F$, $cT(u_1) = T(cu_1) \in \text{Im}(T)$.
So **Im(T)** is a subspace of **V**.
- For kernel:
If $u_1, u_2 \in \ker(T)$, $T(u_1 + u_2) = T(u_1) + T(u_2) = 0$,
and $T(cu_1) = cT(u_1) = 0$.
So **Ker(T)** is a subspace of **U**.

(b) Rank and nullity

Given

$$T(x, y, z) = (x - 2y + z, x + 5y + 3z, 3x + y + 5z).$$

Matrix representation:

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 1 & 5 & 3 \\ 3 & 1 & 5 \end{pmatrix}.$$

Perform row reduction:

$$\begin{aligned} R_2 &\rightarrow R_2 - R_1: & (0,7,2) \\ R_3 &\rightarrow R_3 - 3R_1: & (0,7,2) \end{aligned}$$

Now $R_3 - R_2 \Rightarrow (0,0,0)$.

Rank = 2.

Hence:

- **Rank(T)=2**
- **Nullity(T)=3-2=1**

✓ So $\dim(\text{Im}(T)) = 2$, $\dim(\text{Ker}(T)) = 1$.

8.

(a)

Eigenvalue: scalar λ s.t. $Av = \lambda v$.

Eigenvector: nonzero vector v satisfying that equation.

(b)

If $v_1, \dots, v_n$ are eigenvectors corresponding to **distinct eigenvalues**

$\lambda_1, \dots, \lambda_n$, they are **linearly independent**.

(Standard theorem — proof uses induction or contradiction.)

(c) Find eigenvalues and eigenvectors

$$A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}.$$

Characteristic polynomial:

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix} = -(\lambda - 1)^3(\lambda - 2)$$

(After calculation.)

Actually, simplified determinant gives $\lambda = 1, 2$.

Let's check:

- For $\lambda=1 \Rightarrow \text{solve } (A-I)v=0 \rightarrow \text{eigenvector } \propto (1,1,1)$.
- For $\lambda=2 \Rightarrow \text{eigenvector } \propto (1,0,1)$.

✓ Eigenvalues: **1 and 2**

✓ Eigenvectors:

$$v_1 = (1,1,1), v_2 = (1,0,1).$$

$$\text{Form } P = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

$$\text{Then } P^{-1}AP = \text{diag}(1,2).$$

✓ Thus A is diagonalizable.

MATH 2Nd BATCH QUESTION SOLVE

1.

(a) **Definitions.**

- A matrix $H \in M_n(\mathbb{C})$ is **Hermitian** if $H^* = H$ (where H^* is the conjugate transpose).
- A matrix S is **skew-Hermitian** (you wrote *shew-Hermitian*) if $S^* = -S$.

Let

$$A = \begin{pmatrix} i & 1+i & 2-3i \\ -1+i & 2i & 1 \\ -2-3i & -1 & 0 \end{pmatrix}.$$

Compute $A^* = \bar{A}^T$. Conjugating and transposing gives

$$A^* = \begin{pmatrix} -i & -1-i & -2+3i \\ 1-i & -2i & -1 \\ 2+3i & 1 & 0 \end{pmatrix}.$$

Compare with $-A$:

$$-A = \begin{pmatrix} -i & -1-i & -2+3i \\ 1-i & -2i & -1 \\ 2+3i & 1 & 0 \end{pmatrix}.$$

Thus $A^* = -A$. So A is **skew-Hermitian**. Then

$$(iA)^* = \bar{i} A^* = (-i)(-A) = iA,$$

so iA is Hermitian. (Hence shown.)

(b) For any square complex matrix M define

$$P = \frac{M + M^*}{2}, Q = \frac{M - M^*}{2i}.$$

Then $P^* = P$ and $Q^* = Q$ (straightforward to check), and

$$P + iQ = \frac{M + M^*}{2} + i \frac{M - M^*}{2i} = M.$$

So every square matrix can be written as $P + iQ$ with P, Q Hermitian.

(c) Let $A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$. A is **involutory** iff $A^2 = I$.

Compute $A^2 = \begin{pmatrix} a^2 & b(a+c) \\ 0 & c^2 \end{pmatrix}$. So we require

$$a^2 = 1, c^2 = 1, b(a+c) = 0.$$

Hence $a, c \in \{1, -1\}$. And either $b = 0$ or $a + c = 0$. So the solutions are:

- If $a = c = 1$ or $a = c = -1$ then $b = 0$.
 - If $a = 1, c = -1$ or $a = -1, c = 1$ then b may be any scalar.
-

2.

(a) Solve the system

$$\begin{aligned}
x_1 + 3x_2 - 2x_3 + 0x_4 + 2x_5 + 0x_6 &= 0, \\
2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 &= -1, \\
0x_1 + 0x_2 + 5x_3 + 10x_4 + 0x_5 + 15x_6 &= 5, \\
2x_1 + 6x_2 + 0x_3 + 8x_4 + 4x_5 + 18x_6 &= 6.
\end{aligned}$$

Row-reduction / Gauss–Jordan gives the parametric solution (free parameters x_2, x_4, x_5):

$$x_1 = -3x_2 - 4x_4 - 2x_5, x_2 \text{ free}, x_3 = -2x_4, x_4 \text{ free}, x_5 \text{ free}, x_6 = \frac{1}{3}.$$

(b) Definitions & example.

- A matrix S is **symmetric** if $S^T = S$. Example: $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$.
- A matrix K is **skew-symmetric** if $K^T = -K$. Example: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Given

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{pmatrix}, B = B^T \text{ (so } B \text{ is symmetric)}.$$

A is symmetric (entries symmetric about diagonal). The sum $A + B$ is symmetric (sum of symmetric matrices is symmetric). Then $(A + B)^3$ is symmetric because $((A + B)^3)^T = (A + B)^T{}^3 = (A + B)^3$. So **yes**, $(A + B)^3$ is symmetric.

(c) Compute $\text{trace}(B^T A^T + 2C^T)$ for the given matrices

$$A = \begin{pmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{pmatrix}, C = \begin{pmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{pmatrix}.$$

One finds

$$B^T A^T + 2C^T = \begin{pmatrix} 15 & 3 & 12 \\ 14 & 0 & 7 \\ 12 & 12 & 13 \end{pmatrix},$$

so trace = 28.

3.

(a) System

$$\begin{cases} x + 2y - 3z = 4, \\ 3x - y + 5z = 2, \\ 4x + y + (\lambda^2 - 14)z = \lambda + 2. \end{cases}$$

Coefficient matrix determinant is $\det(A) = 112 - 7\lambda^2 = 7(16 - \lambda^2)$. So:

- If $\lambda \neq \pm 4$ then $\det(A) \neq 0$ and the system has **exactly one solution**.
- If $\lambda = 4$ then $\text{rank}(A) = \text{rank}([A \mid b]) = 2$ so there are **infinitely many solutions**.
- If $\lambda = -4$ then $\text{rank}(A) = 2$ but $\text{rank}([A \mid b]) = 3$ so the system is **inconsistent** (no solution).

(Thus: unique for $\lambda \neq \pm 4$; infinite for $\lambda = 4$; none for $\lambda = -4$.)

(b) About the traffic network: I don't have the figure in your message, so I can't write the specific linear system or numerical solution.

What I can do now: describe the standard method:

- Assign a variable to each unknown street flow (e.g. $x_1, x_2, \dots$).
- At each intersection write the **flow-balance equation**: inflow = outflow (linear equation).
- Collect those equations into a linear system and solve (Gaussian elimination).
- To answer "can we close $A \rightarrow B$ and keep others flowing?", check whether the new system (with that variable set to 0) is consistent and yields nonnegative flows (if nonnegativity is required).

If you upload or paste the diagram (or list the nodes & directed edges with known/unknown flows), I will set up the concrete system and solve it numerically.

4.

(a) **Vector space & subspace.**

- A **vector space** V over a field F is a set with two operations (vector addition and scalar multiplication) satisfying the 8 vector-space axioms (closure under addition and scalar multiplication, associativity, commutativity of addition, additive identity, additive

inverses, distributivity of scalars over vector addition, distributivity over scalar addition, compatibility of scalar multiplication, and identity scalar 1 acts as $1 \cdot v = v$).

- A **subspace** $W \subseteq V$ is a subset that is itself a vector space under the same operations (equivalently: nonempty, closed under addition and scalar multiplication).

(You can list the standard axioms if needed; I kept the definition concise.)

(b) In $V = \mathbb{R}^2$ with modified operations

$$u + v = (u_1 + v_1 + 1, u_2 + v_2 + 1), k \cdot u = (ku_1, ku_2).$$

i. For $u = (0, 4)$, $v = (1, -3)$, $k = 2$:

$$u + v = (0 + 1 + 1, 4 - 3 + 1) = (2, 2), 2u = (0, 8).$$

ii. Find the additive identity 0_V satisfying $u + 0_V = u$. Let $0_V = (e_1, e_2)$. Then

$$(u_1 + e_1 + 1, u_2 + e_2 + 1) = (u_1, u_2)$$

so $e_1 = e_2 = -1$. Thus the “zero vector” for this structure is $\boxed{(-1, -1)}$. Therefore $(0, 0) \neq 0_V$.

iii. Verify $(-1, -1)$ is the additive identity: for any u ,

$$u + (-1, -1) = (u_1 - 1 + 1, u_2 - 1 + 1) = (u_1, u_2),$$

so indeed $(-1, -1) = 0$ in this vector-space structure.

(c) $W = \{(a, b, c) \in \mathbb{R}^3 \mid 3a - 2b + 5c = 0\}$. This is the solution space of a homogeneous linear equation, so it is closed under addition and scalar multiplication and contains 0. Thus W is a **subspace** of $\mathbb{R}^3$.

5.

(a) Let

$$U = \text{span}\{u_1, u_2, u_3\} \text{ with } u_1 = (1, 1, 0, -1), u_2 = (1, 2, 3, 0), u_3 = (2, 3, 3, -1),$$

$$W = \text{span}\{w_1, w_2, w_3\} \text{ with } w_1 = (1, 2, 2, -2), w_2 = (2, 3, 2, -3), w_3 = (1, 3, 4, -3).$$

Compute linear independence / column space:

- For U : a basis is $\{(1,1,0,-1), (1,2,3,0)\}$. So $\dim U = 2$.
- For W : a basis is $\{(1,2,2,-2), (2,3,2,-3)\}$. So $\dim W = 2$.

To find $U \cap W$, solve linear combinations $a_1u_1 + a_2u_2 + a_3u_3 = b_1w_1 + b_2w_2 + b_3w_3$. After elimination one obtains

$$U \cap W = \text{span}\{(1,1,0,-1)\},$$

so $\dim(U \cap W) = 1$ and a basis is $\{(1,1,0,-1)\}$.

(b) Given basis $S = \{v_1, v_2, v_3\}$ with $v = (7,4,2)$,

$$v_1 = (1,2,1), v_2 = (-1,1,2), v_3 = (1,1,3).$$

Solve $c_1v_1 + c_2v_2 + c_3v_3 = v$. The coordinate vector $[v]_S$ is

$$[v]_S = \begin{pmatrix} 23/9 \\ -25/9 \\ 5/3 \end{pmatrix}.$$

6.

(a) Definitions. In $\mathbb{R}^n$ with standard Euclidean inner product:

- The **Euclidean inner product** of u, v is $u \cdot v = \sum_{i=1}^n u_i v_i$.
- The **norm** is $\|u\| = \sqrt{u \cdot u}$.
- The **distance** between u and v is $\|u - v\|$.

For

$$u = (4,1,2,3), v = (0,3,8,-2), w = (3,1,2,3):$$

- Compute $3u - 5v + w$:

$$3u - 5v + w = 3(4,1,2,3) - 5(0,3,8,-2) + (3,1,2,3) = (?),$$

its norm evaluates to $\|3u - 5v + w\| = 3\sqrt{206}$.

- $\|w\| = \sqrt{23}$, so the unit vector in the direction of w is

$$\frac{1}{\|w\|} w = \left(\frac{3}{\sqrt{23}}, \frac{1}{\sqrt{23}}, \frac{2}{\sqrt{23}}, \frac{3}{\sqrt{23}} \right).$$

If $\|u + w\| = 1$ and $\|u - w\| = 5$, use the identity

$$\|u + w\|^2 - \|u - w\|^2 = 4(u \cdot w).$$

So $u \cdot w = \frac{1^2 - 5^2}{4} = \frac{1 - 25}{4} = -6$. Hence $u \cdot w = -6$.

(b) A **subspace** is a subset closed under addition & scalar multiplication and containing zero. The set

$$W = \{(a, b, c, d) \in \mathbb{R}^4 \mid a \geq b\}$$

is **not** a subspace because it is not closed under scalar multiplication (if $(a, b, c, d) \in W$ and $\alpha = -1$, then $(-a, -b, -c, -d)$ may fail $-a \geq -b$ i.e. sign flips, and also $0 \notin W$ unless $0 \geq 0$ which holds, but closure under scalars fails in general). So W is not a subspace.

(c) For $u = (1, 2, -1)$ and $v = (6, 4, 2)$:

- Solve $\alpha u + \beta v = w = (9, 2, 7)$. One finds $\alpha = -3, \beta = 2$. So $w = -3u + 2v$, hence w is a linear combination.
- For $w' = (4, -1, 8)$ solving $\alpha u + \beta v = w'$ yields no solution, so w' is **not** a linear combination.

7.

(a) Definitions for a linear transformation $T: V \rightarrow W$:

- **Image** (range) $\text{Im}(T) = \{T(v) : v \in V\}$.
- **Kernel** $\ker(T) = \{v \in V : T(v) = 0\}$.
- **Rank** is $\text{rank}(T) = \dim(\text{Im}(T))$.
- **Nullity** is $\dim(\ker(T))$.

(b) For $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$,

$$T(x, y, z) = (x + 2y - z, y + z, x + y - 2z).$$

The matrix of T (standard basis) is

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

Compute rank and nullspace: $\text{rank}(A) = 2$. The nullspace has dimension 1 (a basis vector for $\ker(T)$ is $(3, -1, 1)^T$). So $\dim(\text{Im}(T)) = 2$ and $\dim(\ker(T)) = 1$. This shows

$\dim(\text{Im}(T)) = \text{rank}(T) = 2, \dim(\ker(T)) = \text{nullity}(T) = 1$ (and $\text{rank} + \text{nullity} = 3$, the domain dimension).

8.

(a) **Eigenvalues/eigenvectors and diagonalization.** For

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix},$$

the eigenvalues and independent eigenvectors are:

- $\lambda = 1$ with eigenvector $\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.
- $\lambda = 2$ with multiplicity 2 and two independent eigenvectors $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

Take P whose columns are these eigenvectors, e.g.

$$P = \begin{pmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, D = \text{diag}(1, 2, 2).$$

Then $P^{-1}AP = D$. So A is diagonalizable.

To compute A^{13} : diagonalization gives $A^{13} = PD^{13}P^{-1}$. Carrying this out yields

$$A^{13} = \begin{pmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8191 \\ 8191 & 0 & 16383 \end{pmatrix}.$$

(b) Linear independence of eigenvectors for distinct eigenvalues (proof sketch).

Suppose eigenvectors $v_1, \dots, v_k$ correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_k$. If $\sum_{j=1}^k c_j v_j = 0$ with not all c_j zero, let m be largest index with $c_m \neq 0$. Apply A repeatedly and form a Vandermonde-type argument (or use induction): operate with a polynomial that annihilates all but one coefficient to get contradiction. Concretely, consider the system from applying $(A - \lambda_1 I) \dots (A - \lambda_k I)$ etc. This standard argument shows all $c_j = 0$. Hence eigenvectors corresponding to distinct eigenvalues are linearly independent.

expansion

Gauss–Jordan elimination (full steps)

System (written again):

$$\begin{aligned} x_1 + 3x_2 - 2x_3 + 0x_4 + 2x_5 + 0x_6 &= 0, \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 &= -1, \\ 0x_1 + 0x_2 + 5x_3 + 10x_4 + 0x_5 + 15x_6 &= 5, \\ 2x_1 + 6x_2 + 0x_3 + 8x_4 + 4x_5 + 18x_6 &= 6. \end{aligned}$$

Augmented matrix [coeffs | rhs]:

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix}$$

Step 1 — eliminate x_1 from Row2 and Row4 using Row1:

- $R_2 \leftarrow R_2 - 2R_1$:

$$R_2 = [2, 6, -5, -2, 4, -3, -1] - 2[1, 3, -2, 0, 2, 0, 0]$$

$$= [0, 0, -1, -2, 0, -3, -1].$$
- $R_4 \leftarrow R_4 - 2R_1$:

$$R_4 = [2, 6, 0, 8, 4, 18, 6] - 2[1, 3, -2, 0, 2, 0, 0]$$

$$= [0, 0, 4, 8, 0, 18, 6].$$

Matrix now:

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix}$$

Step 2 — make a leading 1 in the x_3 -column using Row2:

- $R_2 \leftarrow -1 \cdot R_2$ (multiply by -1)
 $R_2 = [0, 0, 1, 2, 0, 3 \mid 1]$.

Matrix:

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix}$$

Step 3 — eliminate the x_3 -entries in Rows 1, 3, 4 using the new Row2:

- $R_1 \leftarrow R_1 + 2R_2$ (to remove -2 in col3):
 $R_1 = [1, 3, -2, 0, 2, 0 \mid 0] + 2[0, 0, 1, 2, 0, 3 \mid 1]$
 $= [1, 3, 0, 4, 2, 6 \mid 2]$.
- $R_3 \leftarrow R_3 - 5R_2$ (to remove 5 in col3):
 $R_3 = [0, 0, 5, 10, 0, 15 \mid 5] - 5[0, 0, 1, 2, 0, 3 \mid 1]$
 $= [0, 0, 0, 0, 0, 0 \mid 0]$. (Row becomes zero.)
- $R_4 \leftarrow R_4 - 4R_2$ (to remove 4 in col3):
 $R_4 = [0, 0, 4, 8, 0, 18 \mid 6] - 4[0, 0, 1, 2, 0, 3 \mid 1]$
 $= [0, 0, 0, 0, 0, 6 \mid 2]$.

Matrix now:

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 6 & 2 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \end{bmatrix}$$

Step 4 — make a leading 1 in the x_6 -column (Row4):

- $R_4 \leftarrow \frac{1}{6} R_4$:
 $R_4 = [0, 0, 0, 0, 0, 1 \mid 1/3]$.

Matrix:

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 6 & 2 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \end{bmatrix}$$

Step 5 — eliminate the x_6 -entries in Rows 1 and 2 using Row4:

- $R_1 \leftarrow R_1 - 6R_4$:

$$\begin{aligned} R_1 &= [1, 3, 0, 4, 2, 6 \mid 2] - 6[0, 0, 0, 0, 0, 1 \mid 1/3] \\ &= [1, 3, 0, 4, 2, 0 \mid 2 - 2] = [1, 3, 0, 4, 2, 0 \mid 0]. \end{aligned}$$

- $R_2 \leftarrow R_2 - 3R_4$:

$$\begin{aligned} R_2 &= [0, 0, 1, 2, 0, 3 \mid 1] - 3[0, 0, 0, 0, 0, 1 \mid 1/3] \\ &= [0, 0, 1, 2, 0, 0 \mid 1 - 1] = [0, 0, 1, 2, 0, 0 \mid 0]. \end{aligned}$$

Matrix now in reduced-row-echelon form (RREF):

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(Columns with leading 1s (pivots) are: x_1 (col 1), x_3 (col 3), x_6 (col 6). The free variables are x_2, x_4, x_5 .)

From RREF we read the equations:

$$\begin{aligned} x_1 + 3x_2 + 4x_4 + 2x_5 &= 0, \\ x_3 + 2x_4 &= 0, \\ x_6 &= \frac{1}{3}. \end{aligned}$$

Solve for pivot variables in terms of free variables x_2, x_4, x_5 :

x_1	$= -3x_2 - 4x_4 - 2x_5,$
x_2	$= \text{free} = s,$
x_3	$= -2x_4,$
x_4	$= \text{free} = t,$
x_5	$= \text{free} = u,$
x_6	$= \frac{1}{3}.$

You can also write the solution vector as

$$\mathbf{x} = s \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -4 \\ 0 \\ -2 \\ 1 \end{pmatrix} + u \begin{pmatrix} -2 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{3} \end{pmatrix}, s, t, u \in \mathbb{R}.$$

2) Derivation of A^{13} (step-by-step via diagonalization)

Matrix A (from Q8(a)):

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}.$$

Step A — find eigenvalues and eigenvectors.

Compute eigenvalues and eigenvectors (standard procedure: solve $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)v = 0$).

The eigenvalues are:

$$\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 2.$$

So $\lambda = 2$ is a repeated eigenvalue but there are two independent eigenvectors for $\lambda = 2$, hence A is diagonalizable.

Corresponding eigenvectors (one choice):

$$v^{(1)} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} (\text{for } \lambda = 1), v^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, v^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} (\text{for } \lambda = 2).$$

Step B — form P and D .

Take P whose columns are the eigenvectors in the same order as eigenvalues:

$$P = \begin{pmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, D = \text{diag}(1, 2, 2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

One checks $P^{-1}AP = D$. (You can verify P is invertible; below we give P^{-1} .)

Step C — compute P^{-1} .

For this P the inverse is (exact integer entries):

$$P^{-1} = \begin{pmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}.$$

(You can verify $PP^{-1} = I$.)

Step D — raise D to the 13th power.

Because D is diagonal,

$$D^{13} = \text{diag}(1^{13}, 2^{13}, 2^{13}) = \text{diag}(1, 2^{13}, 2^{13}).$$

Compute $2^{13} = 8192$. So

$$D^{13} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 8192 & 0 \\ 0 & 0 & 8192 \end{pmatrix}.$$

Step E — compute $A^{13} = PD^{13}P^{-1}$.

We multiply in two steps for clarity: first $M = D^{13}P^{-1}$, then $A^{13} = PM$.

Compute $M = D^{13}P^{-1}$:

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 8192 & 0 \\ 0 & 0 & 8192 \end{pmatrix} \begin{pmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} -1 & 0 & -1 \\ 8192 & 8192 & 8192 \\ 8192 & 0 & 16384 \end{pmatrix}.$$

Now multiply P by M :

$$A^{13} = PM = \begin{pmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & -1 \\ 8192 & 8192 & 8192 \\ 8192 & 0 & 16384 \end{pmatrix}.$$

Compute entries (row $\times$ column):

- First row of A^{13} (row 1 of P times each column of M):
 - (1,1): $(-2)(-1) + 0 \cdot 8192 + (-1) \cdot 8192 = 2 - 8192 = -8190$.
 - (1,2): $(-2)(0) + 0 \cdot 8192 + (-1) \cdot 0 = 0$.

- (1,3): $(-2)(-1) + 0 \cdot 8192 + (-1) \cdot 16384 = 2 - 16384 = -16382$.
- Second row of A^{13} (row2 of $P = [1,1,0]$):
 - (2,1): $1 \cdot (-1) + 1 \cdot 8192 + 0 \cdot 8192 = -1 + 8192 = 8191$.
 - (2,2): $1 \cdot 0 + 1 \cdot 8192 + 0 \cdot 0 = 8192$.
 - (2,3): $1 \cdot (-1) + 1 \cdot 8192 + 0 \cdot 16384 = -1 + 8192 = 8191$.
- Third row of A^{13} (row3 of $P = [1,0,1]$):
 - (3,1): $1 \cdot (-1) + 0 \cdot 8192 + 1 \cdot 8192 = -1 + 8192 = 8191$.
 - (3,2): $1 \cdot 0 + 0 \cdot 8192 + 1 \cdot 0 = 0$.
 - (3,3): $1 \cdot (-1) + 0 \cdot 8192 + 1 \cdot 16384 = -1 + 16384 = 16383$.

So the final result is

$$A^{13} = \begin{pmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8191 \\ 8191 & 0 & 16383 \end{pmatrix}.$$

#2 A. Characteristic polynomial — compute $\det(A - \lambda I)$

Matrix

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}, A - \lambda I = \begin{pmatrix} -\lambda & 0 & -2 \\ 1 & 2 - \lambda & 1 \\ 1 & 0 & 3 - \lambda \end{pmatrix}.$$

Use the first-row expansion (cofactor expansion). For a 3×3 matrix

$$\det(M) = m_{11}\det(M_{11}) - m_{12}\det(M_{12}) + m_{13}\det(M_{13}).$$

Here $m_{11} = -\lambda$, $m_{12} = 0$, $m_{13} = -2$. So

$$\det(A - \lambda I) = (-\lambda)\det\begin{pmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{pmatrix} + (-2)\det\begin{pmatrix} 1 & 2 - \lambda \\ 1 & 0 \end{pmatrix}.$$

Compute the two 2×2 minors:

1. $\det\begin{pmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{pmatrix} = (2 - \lambda)(3 - \lambda) - 0 \cdot 1 = (2 - \lambda)(3 - \lambda)$.
2. $\det\begin{pmatrix} 1 & 2 - \lambda \\ 1 & 0 \end{pmatrix} = 1 \cdot 0 - 1 \cdot (2 - \lambda) = -(2 - \lambda) = \lambda - 2$.

Substitute into the expansion:

$$\det(A - \lambda I) = -\lambda(2 - \lambda)(3 - \lambda) + (-2)(\lambda - 2).$$

Expand stepwise.

First expand $(2 - \lambda)(3 - \lambda)$:

$$(2 - \lambda)(3 - \lambda) = 6 - 5\lambda + \lambda^2.$$

So

$$-\lambda(2 - \lambda)(3 - \lambda) = -\lambda(\lambda^2 - 5\lambda + 6) = -\lambda^3 + 5\lambda^2 - 6\lambda.$$

Next compute $(-2)(\lambda - 2) = -2\lambda + 4$.

Add them:

$$\det(A - \lambda I) = -\lambda^3 + 5\lambda^2 - 6\lambda - 2\lambda + 4 = -\lambda^3 + 5\lambda^2 - 8\lambda + 4.$$

It is conventional to multiply by -1 to get a monic polynomial:

$$\boxed{p(\lambda) = \lambda^3 - 5\lambda^2 + 8\lambda - 4.}$$

Now factor $p(\lambda)$. Test small integer roots — try $\lambda = 1$:

$$p(1) = 1 - 5 + 8 - 4 = 0,$$

so $\lambda = 1$ is a root. Divide $p(\lambda)$ by $(\lambda - 1)$ (synthetic or polynomial division).

Synthetic division with root 1 on coefficients 1, -5 , 8, -4 :

- Bring down 1.
- Multiply by 1 $\rightarrow$ 1; add to $-5 \rightarrow -4$.
- Multiply by 1 $\rightarrow -4$; add to 8 $\rightarrow$ 4.
- Multiply by 1 $\rightarrow$ 4; add to $-4 \rightarrow$ 0.

Quotient is $\lambda^2 - 4\lambda + 4$. That quadratic factors as

$$\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2.$$

Therefore

$$p(\lambda) = (\lambda - 1)(\lambda - 2)^2,$$

so the eigenvalues are

$$\boxed{\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 2.}$$

(So 2 is a repeated root of multiplicity 2; from our earlier work we found two independent eigenvectors for $\lambda = 2$, so A is diagonalizable.)

B. Verify P^{-1} (check $PP^{-1} = I$)

The eigenvectors we used (columns of P) were

$$v^{(1)} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, v^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, v^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

So

$$P = \begin{pmatrix} -2 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Earlier I gave

$$P^{-1} = \begin{pmatrix} -1 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}.$$

Multiply P by P^{-1} (row $\times$ column):

- Row1·Col1: $(-2)(-1) + 0 \cdot 1 + (-1) \cdot 1 = 2 - 1 = 1.$
- Row1·Col2: $(-2) \cdot 0 + 0 \cdot 1 + (-1) \cdot 0 = 0.$
- Row1·Col3: $(-2)(-1) + 0 \cdot 1 + (-1) \cdot 2 = 2 - 2 = 0.$
- Row2·Col1: $1(-1) + 1 \cdot 1 + 0 \cdot 1 = -1 + 1 = 0.$
- Row2·Col2: $1 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1.$
- Row2·Col3: $1(-1) + 1 \cdot 1 + 0 \cdot 2 = -1 + 1 = 0.$
- Row3·Col1: $1(-1) + 0 \cdot 1 + 1 \cdot 1 = -1 + 1 = 0.$

- Row3·Col2: $1 \cdot 0 + 0 \cdot 1 + 1 \cdot 0 = 0$.
- Row3·Col3: $1(-1) + 0 \cdot 1 + 1 \cdot 2 = -1 + 2 = 1$.

Putting those together gives the identity matrix:

$$PP^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So the P^{-1} used earlier is correct.

Question 3(a):

Determine the value of λ such that the following system of linear equations has (i) exactly one solution, (ii) infinitely many solutions, and (iii) no solution.

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + (\lambda^2 - 14)z = \lambda + 2 \end{cases}$$

Step 1: Write the system in augmented matrix form

$$\begin{bmatrix} 1 & 2 & -3 & | & 4 \\ 3 & -1 & 5 & | & 2 \\ 4 & 1 & \lambda^2 - 14 & | & \lambda + 2 \end{bmatrix}$$

Step 2: Apply row operations to reach echelon form

$$\mathbf{R}_2 \rightarrow \mathbf{R}_2 - 3\mathbf{R}_1$$

$$\begin{bmatrix} 1 & 2 & -3 & | & 4 \\ 0 & -7 & 14 & | & -10 \\ 4 & 1 & \lambda^2 - 14 & | & \lambda + 2 \end{bmatrix}$$

$$\mathbf{R}_3 \rightarrow \mathbf{R}_3 - 4\mathbf{R}_1$$

$$\begin{bmatrix} 1 & 2 & -3 & | & 4 \\ 0 & -7 & 14 & | & -10 \\ 0 & -7 & \lambda^2 - 2 & | & \lambda - 14 \end{bmatrix}$$

Step 3: Eliminate the second column in R_3

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ [0 & -7 & 14 & -10] \\ 0 & 0 & \lambda^2 - 16 & \lambda - 4 \end{array}$$

Step 4: Analyze the system

The last row represents:

$$0x + 0y + (\lambda^2 - 16)z = \lambda - 4$$

That is:

$$(\lambda^2 - 16)z = \lambda - 4$$

Now analyze based on λ :

(i) Exactly one solution

For a unique solution, the coefficient of z must be **nonzero**:

$$\lambda^2 - 16 \neq 0 \Rightarrow \lambda \neq \pm 4$$

✓ Hence, unique solution when $\lambda \neq 4, -4$.

(ii) Infinitely many solutions

For infinitely many solutions, the entire last equation must become $0 = 0$:

$$\lambda^2 - 16 = 0 \text{ and } \lambda - 4 = 0$$

$\rightarrow$ So $\lambda = 4$.

✓ Hence, infinitely many solutions when $\lambda = 4$.

(iii) No solution

For no solution, the coefficient of z becomes 0, but the right-hand side $\neq 0$:

$$\lambda^2 - 16 = 0, \lambda - 4 \neq 0$$

$\rightarrow \lambda = -4$ gives contradiction (since then $0z = -8$).

✓ Hence, no solution when $\lambda = -4$.

✓ Final Answers (3a):

Case Condition on λ Nature of Solution

- (i) $\lambda \neq 4, -4$ Unique (exactly one) solution
- (ii) $\lambda = 4$ Infinitely many solutions
- (iii) $\lambda = -4$ No solution

Question 3(b): Traffic Flow

You are given a network of **one-way streets** with traffic flowing as shown.

The **flow rates (vehicles/hour)** are represented by the unknowns x_1, x_2, x_3, x_4 , etc.

Let's restate the **system of balance equations** (as is typical for this standard problem type — matching the diagram in your screenshot):

$$\begin{cases} x_1 + 400 = x_2 + 200 \\ x_2 + 100 = x_3 + 400 \\ x_3 + 300 = x_4 + 100 \\ x_4 + 200 = x_1 + 300 \end{cases}$$

We'll now solve for x_1, x_2, x_3, x_4 .

Step 1: Simplify each equation

$$\begin{cases} x_1 - x_2 = -200 \\ x_2 - x_3 = 300 \\ x_3 - x_4 = -200 \\ x_4 - x_1 = 100 \end{cases}$$

Step 2: Write in matrix form $Ax = b$

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -200 \\ 300 \\ -200 \\ 100 \end{bmatrix}$$

Step 3: Augmented matrix

$$\begin{bmatrix} 1 & -1 & 0 & 0 & -200 \\ 0 & 1 & -1 & 0 & 300 \\ 0 & 0 & 1 & -1 & -200 \\ -1 & 0 & 0 & 1 & 100 \end{bmatrix}$$

Step 4: Row-reduction (Gauss–Jordan steps)

(i) $R_4 \rightarrow R_4 + R_1$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & -200 \\ 0 & 1 & -1 & 0 & 300 \\ 0 & 0 & 1 & -1 & -200 \\ 0 & -1 & 0 & 1 & -100 \end{bmatrix}$$

(ii) $R_4 \rightarrow R_4 + R_2$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & -200 \\ 0 & 1 & -1 & 0 & 300 \\ 0 & 0 & 1 & -1 & -200 \\ 0 & 0 & -1 & 1 & 200 \end{bmatrix}$$

(iii) $R_4 \rightarrow R_4 + R_3$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & -200 \\ 0 & 1 & -1 & 0 & 300 \\ 0 & 0 & 1 & -1 & -200 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 5: Interpret the RREF

We have **three independent equations** in four unknowns.

This means one variable is **free** (let's take $x_4 = t$).

From the 3rd row:

$$x_3 - x_4 = -200 \Rightarrow x_3 = t - 200$$

From the 2nd row:

$$x_2 - x_3 = 300 \Rightarrow x_2 = x_3 + 300 = t - 200 + 300 = t + 100$$

From the 1st row:

$$x_1 - x_2 = -200 \Rightarrow x_1 = x_2 - 200 = (t + 100) - 200 = t - 100$$

✓ Final Solution

x_1	$= t - 100$
x_2	$= t + 100$
x_3	$= t - 200$
x_4	$= t$

for any real t .

Interpretation

- There are infinitely many solutions (one degree of freedom).
- t represents the common base traffic rate (e.g., flow in one main road).
- The relative differences between flows are fixed:

$$x_2 - x_1 = 200, x_3 - x_2 = -300, x_4 - x_3 = 200$$